ANDERSON SERANGOON JUNIOR COLLEGE

JC1 Promotional Examination 2022

GENERAL PAPER	**8807/02**
PAPER 2	**23 September 2022**
INSERT	**1 hour 30 minutes**

READ THESE INSTRUCTIONS FIRST

This Insert contains the passage for Paper 2.

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Youdi Beshem talks about the benefits and harms of Smart Transportation today.

1 Being constantly on the move is part of being human. For our nomadic ancestors, mobility was a matter of survival. But even as our pre-historic predecessors moved from hunter-gatherer lifestyles to form permanent settlements, man's desire to traverse the earth and move to new locations remained. Evidence of this desire was unearthed from millennia ago: the Pesse canoe, made by hollowing out a tree trunk, is the oldest boat ever found, dating as far back as 7600 BC. In 3500 BC, the wooden wheel was invented in ancient Iraqi civilisation, giving rise to transportation methods hardly dreamt of then. It was in the last 400 years, however, that developments in transportation moved at a dizzying pace, bringing us more innovations in transport – from horses and steam trains to automobiles, and from moving across land and water to air and space!

2 Today, with the emergence of intelligent transportation systems, societies have entered the long-awaited and inevitable stage of Smart Transportation. With advancing communication technologies, automation and high-speed networks, cities have revolutionised mobility for cities and its inhabitants. And if these are not impressive enough, examples of transportation technology that until recently only existed in science fiction is now the stuff of reality**:** in 2022, a flying car capable of hitting speeds over 160kmh and altitudes above 2,500m was actually issued a certificate of airworthiness by the Slovak Transport Authority!

3 The benefits of Smart Transportation are inexhaustible as countries constantly upgrade applications of modern technology and management strategies by making use of data collected from commuters and the network. Satellite navigation allows driving routes to be calculated effortlessly. Traffic lights have been developed to monitor surroundings in real-time and adjust to current traffic conditions. If you drive home at rush hour and see green all the way, it means dynamic signals have turned traffic lights green to maintain traffic flow. Dublin is one such example where a city-wide sensor system was implemented to avoid the unnecessary slowing down of vehicle flow. Furthermore, Smart Transportation has seen the use of navigation to find multiple possible routes to a destination plus the ability to alert drivers quickly of potentially hazardous situations. Having these choices allows commuters to avoid the daily grind (liberating us from at least one form of boredom in our lives!). Combining machine-learning with the Internet of Things and 5G, autonomous transportation systems are intelligent enough to predict possible traffic situations and reduce the frequency of accidents.

4 With the elimination of human factors such as distraction, fatigue and emotions, increased cost savings in repairs are possible. Autonomous transportation of goods is a lifesaver as it enables supply chains to continue to operate despite unforeseen circumstances such as the Coronavirus (Covid-19) pandemic. Crises such as the above have proven that the world's supply chains are vulnerable to disruption, as unwell workers or drivers travelling from one region to another become a public health hazard. Projects such as Wyoming's connected vehicle project might be the key to forming an autonomous supply chain powered by Smart Transport and logistics systems to move critical goods such as food and emergency supplies without the need to risk human drivers. With its intricately connected network, valuable data can be easily accessible to enable efficient dispatch of resources and manpower in response to emergencies. Cost savings also applies to riders when inexpensive public transit can compete with private vehicle ownership.

5 Other noteworthy benefits of intelligent transport lie at the heart of Seoul's metro systems which provide Smart payment systems, 4G, and Wi-Fi services. Think Smart payment and hassle-free cashless payment – buzzword of the 21st century – comes to mind! The latter two are a breath of relief for commuters as they now can send that text message to a friend explaining why they are running late for their appointment. This is a godsend on top of being able to reply to an urgent work email while enroute home. Such communication, especially underground, is not a guaranteed privilege in other countries. Seoul's subways are also climate controlled and have seat warmers. What cosier way is there to travel during wintry seasons? Their stations also have digital terminals so commuters can look up the best route for their journey, which is a practical use of technology and space within their station - and a real people pleaser! Smart Transport now even features the convenience of virtual grocery stores in some places - walls that look like store shelves, displaying product images and prices, each with a QR code. Consumers scan products and orders are delivered within the day. Perhaps the most critical advantage, is the safety-enhancing aspect of Smart Transport systems in an era where transport is often a target for individuals seeking to spread terror. In Atlanta, USA, the police introduced a free app, designed for people to report suspicious activity they may witness or be involved in on public transport. It is specially designed for areas that have poor cell phone signals like the subway underground. It enables two-way communication between the user and the police, easily and quickly in order to respond to emergency situations.

6 The plethora of benefits, however, is sometimes a cloak that covers the subtle harms that accompany innovations like Smart Transportation. Often, these problems come to the surface when Smart city transportation systems are implemented, and centre around power consumption and responsible data management. Smart cities depend on a massive number of sensors which put an immense strain on our already diminishing power sources. Another evil which spreads its tentacles wide, causing great harm, is the compromising of personal data online. Transportation in Smart cities operate based on data, which is the essence and the indisputable life force of such technologically advanced states. While much of the information needed is anonymous, people will need to have a mental and behavioural shift as they get used to sensors in a city collecting the signals that a smartphone emits throughout the day. No doubt, we need responsible laws and policies to manage the data that is collected in order for Smart cities to thrive in the future.

7 Another major fear among sceptics is the vulnerability of Smart Transport to cyber-attacks. It has always been painfully obvious that transportation is a core tenet of any nation to function and nations around the world are bracing themselves for the next criminal who will outsmart their Smart Transport systems. Truth be told, the attacks are nothing new—only the tools are novel. Physical threats include criminals stealing cars, terrorists using vehicles as weapons (like the 2016 lorry attack in France), and bad actors holding public transportation hostage. In these cases, the threats were immediately identified and neutralised. But cyber-attacks are far more insidious and isolating the source of the malice itself is a huge task, let alone responding to it.

8 The ongoing Covid-19 pandemic has brought our previously nonstop movement to a palpable halt. With people purely working from home, abandoned offices almost painted a bleak dystopian vignette of a society devoid of face-to-face interaction – a scene which is slowly dissolving. In fact, transport developments are steadily gathering pace again. Multinational corporations have started calling workers back to the office. Far from pulling the plug on fast-paced transportation, the Covid-19 pandemic has heightened the need for a safer and more efficient transport system. Furthermore, a report from C40's Cities Climate Leadership Group asserts that in a world reeling from the impact of Covid-19, investing in public transport could create 4.6 million jobs by 2030. The scampering to build more robust transport systems is a testimony of how much we detested being cooped up at home during the Covid-19 lockdowns, and of a deeper fundamental truth: freedom of movement is truly a basic human need.

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